# CHAPTER 1

# INTRODUCTION

## 1.1 Introduction

Road accidents remain one of the leading causes of fatalities and injuries worldwide, with a significant proportion attributable to delayed driver response and the absence of real-time situational awareness between vehicles. The emergence of Intelligent Transportation Systems (ITS) has introduced the concept of connected vehicles, where automobiles are active participants in a wider communication ecosystem. Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication enable real-time exchange of vehicle data including speed, location, heading, and motion parameters between nearby vehicles and roadside units, facilitating timely collision warnings and adaptive traffic management.

Existing Advanced Driver Assistance Systems (ADAS) rely on onboard sensors such as cameras, radar, and ultrasonic transducers. While effective within their line-of-sight, these systems cannot detect hazards around sharp curves, behind large vehicles, or in adverse weather conditions. Wireless vehicular communication directly addresses this gap by extending situational awareness beyond the physical sensor range of any individual vehicle.

This project develops a Prototype Framework for V2V and V2I Communication for Collision Avoidance using embedded hardware. The system employs an STM32F411 Nucleo-64 microcontroller integrated with a GPS Neo-6M module, MPU6050 inertial sensor, OLED display, and interchangeable wireless modules — XBee, NRF24L01+, and LoRa. NS-3 network simulations of DSRC, 4G LTE, and 5G NR environments are additionally used to evaluate and compare communication standards for intelligent transportation applications.

## 1.2 Motivation

Rapid urbanization has increased vehicle density on roads, leading to severe traffic congestion and higher accident rates. Despite advances in individual vehicle sensing, inter-vehicle communication remains absent from affordable prototype platforms. The absence of a unified, low-cost embedded V2V prototype that simultaneously evaluates multiple wireless communication technologies motivates this work. The comparison between practical hardware-based protocols (XBee, NRF24L01+, LoRa) and simulated industrial standards (DSRC, 4G LTE, 5G NR via NS-3) fills a critical gap in the literature, providing empirical data to guide future ITS deployments.

## 1.3 Objectives

The objectives of this project are:

- To design and develop a prototype framework for Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication using STM32, GPS, and wireless communication modules for collision avoidance.
- To evaluate and compare the performance of different communication protocols such as XBee, NRF24L01+, and LoRa, along with simulated DSRC, 4G, and 5G networks using NS-3 for intelligent transportation applications.

## 1.4 Outcomes

The expected outcomes of this project are:

- A functional prototype for Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication is developed using STM32, GPS Neo-6M, MPU6050 sensor, and wireless communication modules.
- Comparative analysis of XBee and NRF24L01+ communication protocols, along with NS-3 simulation of DSRC, 4G, and 5G networks, is successfully performed to evaluate communication performance.

## 1.5 Social Relevance

The lack of real-time inter-vehicle communication and delayed driver response are major contributing factors to road accidents, which continue to rank among the top causes of fatalities and injuries globally. By creating a prototype V2V and V2I communication architecture that broadcasts vehicle location, speed, and motion data to neighboring vehicles and roadside infrastructure in real time, this project directly addresses this pressing public safety issue. The system has the potential to lower accident rates on highway and urban roads by facilitating prompt collision alerts and situational awareness. This work directly relates to the national goal of reducing road fatalities and enhancing transportation safety.

## 1.6 Problem Statement

Existing transportation systems lack a reliable, low-latency inter-vehicle communication mechanism capable of sharing real-time vehicle data across varying wireless technologies and network standards, thereby limiting collision avoidance and intelligent transportation applications.

## 1.7 Organization of the Report

This report is organized into five chapters. Chapter 1 introduces the project motivation, objectives, outcomes, and social relevance. Chapter 2 presents a literature survey of existing V2X communication technologies and identifies the research gap. Chapter 3 describes the complete system design including the block diagram, hardware specifications, and software architecture. Chapter 4 covers hardware testing, simulation results, and experimental observations. Chapter 5 presents result analysis and conclusions.

# CHAPTER 2

# LITERATURE SURVEY

## 2.1 Introduction

The rapid growth of the automotive industry and the increasing number of vehicles on roads have significantly improved transportation efficiency. However, this growth has also resulted in higher traffic congestion, road accidents, fuel consumption, and environmental pollution. According to the World Health Organization (WHO), road traffic accidents claim approximately 1.19 million lives every year, making them one of the leading causes of death worldwide, particularly among young adults. Most accidents occur due to human errors such as distracted driving, over-speeding, delayed reaction, improper lane changes, and failure to detect vehicles in blind spots. These challenges have encouraged researchers and automobile manufacturers to develop intelligent transportation technologies capable of improving road safety and traffic management.

Intelligent Transportation Systems (ITS) have emerged as an effective solution to overcome the limitations of conventional transportation systems. ITS integrates advanced sensing, wireless communication, computing, and networking technologies to enable cooperation among vehicles, roadside infrastructure, traffic management centers, and cloud services. Unlike traditional systems where each vehicle relies solely on its onboard sensors such as cameras, RADAR, and LiDAR, ITS allows vehicles to exchange information with surrounding entities in real time. This cooperative communication enables vehicles to become aware of hazards beyond their direct line of sight, thereby significantly improving driving safety and operational efficiency.

Vehicle-to-Everything (V2X) communication forms the foundation of modern Intelligent Transportation Systems. V2X enables the exchange of real-time information between vehicles, roadside infrastructure, pedestrians, cellular networks, and cloud servers. The communication allows vehicles to broadcast critical information such as vehicle position, speed, acceleration, braking status, road conditions, and traffic events. The timely exchange of this information enables applications including forward collision warning, blind spot detection, emergency electronic brake lights, cooperative adaptive cruise control, lane change assistance, intersection collision avoidance, emergency vehicle prioritization, and autonomous driving support.

Among the various V2X communication modes, Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communications are considered the most critical for road safety applications. V2V communication allows direct exchange of safety messages between nearby vehicles without requiring centralized infrastructure. In contrast, V2I communication enables vehicles to interact with Road Side Units (RSUs), traffic signals, parking infrastructure, toll plazas, and traffic management systems. Together, these communication mechanisms improve situational awareness, reduce communication delays, and support intelligent traffic management.

Several wireless communication technologies have been proposed for implementing V2X communication, including Dedicated Short-Range Communication (DSRC), Cellular Vehicle-to-Everything (C-V2X), LTE-V2X, 5G New Radio (NR-V2X), Wi-Fi, ZigBee, Bluetooth, LoRa, and proprietary radio-frequency modules such as the NRF24L01+. Each technology offers unique advantages in terms of communication range, latency, throughput, infrastructure requirements, energy consumption, and deployment cost. Selecting an appropriate communication technology depends on the target application, traffic density, communication distance, reliability requirements, and economic feasibility.

In recent years, the automotive industry has witnessed significant advancements in connected and autonomous vehicle technologies. Organizations such as IEEE, the 3rd Generation Partnership Project (3GPP), the European Telecommunications Standards Institute (ETSI), the Society of Automotive Engineers (SAE), and the 5G Automotive Association (5GAA) have developed standardized communication protocols to ensure interoperability and reliability among connected vehicles. These standards continue to evolve with the introduction of IEEE 802.11bd, Cellular V2X Release 16 and Release 17, and 5G NR-V2X technologies, which provide ultra-low latency, high reliability, and enhanced scalability for next-generation transportation systems.

Despite these advancements, several practical challenges remain. Existing commercial V2X systems often depend on expensive communication infrastructure, high deployment costs, continuous cellular connectivity, and complex network architectures. Such requirements make their deployment difficult in rural areas, developing countries, and low-cost transportation applications. Furthermore, there is limited comparative research on low-cost embedded wireless technologies that can provide localized V2V and V2I communication without relying on cellular infrastructure.

To address these limitations, this project investigates the implementation of a modular V2X communication platform based on the STM32 microcontroller using multiple wireless technologies including NRF24L01+, XBee, and LoRa modules. The system focuses on collision avoidance and blind spot detection by exchanging real-time safety messages between vehicles and roadside units. A comparative evaluation of these wireless technologies is performed using both embedded hardware implementation and NS-3 network simulation to analyze communication performance under different traffic conditions. The parameters considered include Packet Delivery Ratio (PDR), latency, throughput, packet loss, communication range, and power consumption.

This chapter presents a comprehensive review of the existing literature related to V2V and V2I communication technologies. It begins with the historical evolution and necessity of V2X communication, followed by the fundamental concepts of various V2X communication modes. The chapter further discusses the performance metrics used to evaluate vehicular communication systems, reviews recent research contributions from leading researchers, presents current industrial developments and market trends, identifies the existing research gaps, and finally establishes the motivation for the proposed work.

## 2.2 History and Need of V2X Communication

### 2.2.1 Need for V2X

Despite continuous improvements in vehicle safety systems, road traffic accidents remain one of the major global public safety concerns. According to the World Health Organization, nearly 1.19 million people lose their lives every year due to road traffic accidents, while millions more suffer permanent injuries. Human error contributes to more than 90% of these accidents, highlighting the limitations of relying solely on driver perception and decision-making.

Modern vehicles are equipped with advanced driver assistance systems (ADAS) such as cameras, RADAR, ultrasonic sensors, and LiDAR. Although these sensors significantly improve vehicle perception, they are constrained by line-of-sight limitations. Buildings, large vehicles, sharp road curves, adverse weather conditions, fog, heavy rain, and blind spots can obstruct sensor visibility, preventing timely detection of hazards.

Vehicle-to-Everything communication addresses these limitations by enabling vehicles to exchange information beyond the visible environment. Through continuous wireless communication, vehicles become aware of neighboring vehicles, traffic signals, roadside infrastructure, pedestrians, and traffic management systems. This cooperative awareness allows drivers and autonomous driving systems to anticipate dangerous situations before they become visible, thereby reducing reaction time and improving road safety.

V2X communication is particularly important for applications such as forward collision warning, blind spot detection, emergency electronic brake lights, cooperative adaptive cruise control, lane merge assistance, intersection collision avoidance, emergency vehicle prioritization, intelligent traffic signal control, and autonomous vehicle coordination. The technology also contributes to smoother traffic flow, reduced travel time, lower fuel consumption, decreased carbon emissions, and improved utilization of transportation infrastructure.

With the rapid advancement of smart cities, connected vehicles, electric vehicles, and autonomous driving technologies, V2X communication has become a fundamental enabling technology for future transportation systems. Governments, automobile manufacturers, telecommunication companies, and research organizations worldwide continue to invest heavily in V2X research to improve road safety, traffic efficiency, and sustainable urban mobility.

### 2.2.2 Evolution of Vehicle-to-Everything Communication

Transportation has played a vital role in the economic and social development of civilizations for centuries. The rapid increase in urbanization and population over the last few decades has led to a substantial rise in the number of vehicles operating on road networks. Although modern vehicles have become safer and more efficient, traffic congestion, road accidents, fuel consumption, and environmental pollution have also increased significantly. Conventional transportation systems primarily rely on human decision-making, visual observation, and static traffic control mechanisms, which often fail to respond effectively to dynamic traffic conditions.

The concept of Intelligent Transportation Systems (ITS) emerged during the late 1980s and early 1990s as researchers began integrating communication technologies, computing systems, and automation into transportation infrastructure. Early ITS applications mainly focused on electronic toll collection, traffic monitoring, and intelligent traffic signal control. These systems gradually evolved with the introduction of Global Positioning System (GPS) technology, Geographic Information Systems (GIS), wireless communication networks, and embedded microcontrollers.

As wireless communication technologies advanced, researchers proposed Vehicular Ad Hoc Networks (VANETs), a specialized form of Mobile Ad Hoc Networks (MANETs), in which vehicles communicate directly with one another while continuously changing their positions. VANETs formed the technological foundation for modern Vehicle-to-Everything communication by enabling decentralized communication among moving vehicles without permanent network infrastructure.

To support standardized vehicular communication, the IEEE introduced the IEEE 802.11p amendment, popularly known as Dedicated Short-Range Communication (DSRC). DSRC operates in the 5.9 GHz licensed spectrum and was specifically designed for low-latency safety-critical communication between vehicles and roadside infrastructure. It enables direct exchange of Basic Safety Messages (BSMs) with communication delays typically below 100 milliseconds, making it suitable for emergency braking alerts, collision avoidance, and cooperative driving applications.

Although DSRC demonstrated excellent performance in short-range communication, its deployment required dedicated roadside infrastructure and specialized onboard communication units, resulting in high installation costs. To overcome these limitations, the 3rd Generation Partnership Project (3GPP) introduced Cellular Vehicle-to-Everything (C-V2X) communication in Release 14. Unlike DSRC, C-V2X utilizes existing cellular infrastructure while also supporting direct communication between vehicles through the PC5 side link interface. This approach improved communication range, reliability, and scalability while enabling integration with cloud services and intelligent traffic management systems.

The evolution continued with LTE-V2X and later 5G New Radio Vehicle-to-Everything (5G NR-V2X). 5G NR-V2X provides ultra-reliable low-latency communication (URLLC), higher throughput, network slicing, edge computing integration, and enhanced support for autonomous driving applications. The technology enables communication delays below 10 milliseconds and supports advanced cooperative perception, platooning, remote driving, and real-time high-definition map sharing.

Currently, researchers are exploring sixth-generation (6G) communication technologies that integrate artificial intelligence, terahertz communication, integrated sensing and communication (ISAC), digital twins, satellite communication, and edge intelligence into vehicular networks. These developments are expected to transform future transportation systems into fully autonomous, cooperative, and self-optimizing mobility ecosystems.

## 2.3 Fundamentals of Vehicle-to-Everything (V2X) Communication

Vehicle-to-Everything (V2X) communication is a communication framework that enables vehicles to exchange information with surrounding vehicles, roadside infrastructure, pedestrians, communication networks, and cloud servers. Unlike conventional vehicles that rely solely on onboard sensors such as cameras, RADAR, LiDAR, and ultrasonic sensors, V2X extends the perception capability of vehicles beyond the driver's line of sight. By sharing real-time information regarding vehicle position, speed, acceleration, braking status, road conditions, and traffic events, V2X enables cooperative driving and intelligent transportation.

The term "Vehicle-to-Everything" encompasses multiple communication modes, each designed to address specific transportation challenges. These include Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Network (V2N), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Grid (V2G). Together, these communication mechanisms create an interconnected transportation ecosystem that improves safety, traffic efficiency, environmental sustainability, and driving comfort. transportation planning.

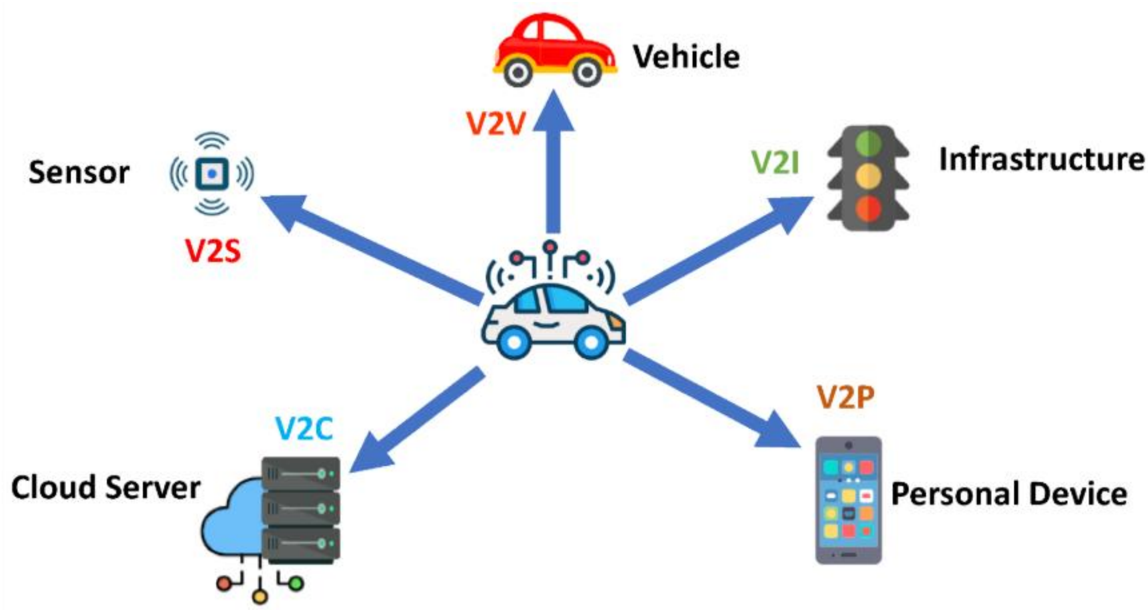



**Figure 2.1** Vehicle-to-Everything communication architecture [18]

The primary objective of V2X communication is to provide timely and reliable dissemination of safety-critical information. Figure 2.1 shows many road accidents occur due to delayed driver reactions or limited visibility, V2X enables vehicles to receive hazard warnings several hundred milliseconds before the driver can visually detect the danger. This additional reaction time significantly reduces collision probability and enhances overall road safety.

### 2.3.1 Overall V2X Communication Architecture

A complete V2X communication environment consists of multiple interconnected entities working together to improve transportation safety and efficiency.

The major components include:

- On Board Unit (OBU)
- Road Side Unit (RSU)
- GPS Satellite System
- Traffic Management Center
- Cellular Network
- Cloud Server
- Edge Computing Nodes

- Emergency Services
- Smart Traffic Lights
- Connected Vehicles

The On-Board Unit installed inside each vehicle continuously collects information from GPS receivers, inertial sensors, cameras, and vehicle control units. The collected data is transmitted through wireless communication technologies to nearby vehicles, RSUs, and cloud servers. Road Side Units perform local processing and forward aggregated traffic information to traffic management centers. Cloud platforms analyze large-scale transportation data using artificial intelligence and distribute optimized routing decisions back to connected vehicles.

This distributed communication architecture enables cooperative perception, predictive traffic management, collision avoidance, autonomous driving support, and intelligent

### 2.3.2 Vehicle-to-Vehicle (V2V) Communication

Vehicle-to-Vehicle (V2V) communication refers to the direct wireless exchange of information between nearby vehicles without requiring any centralized communication infrastructure. Each vehicle continuously broadcasts its dynamic status, including geographical location, vehicle speed, acceleration, heading direction, braking status, and steering information. Neighboring vehicles receive these messages and process them to predict potential collision scenarios or hazardous driving conditions.

The communication generally occurs using Dedicated Short-Range Communication (DSRC), Cellular Vehicle-to-Everything (C-V2X), IEEE 802.11p, IEEE 802.11bd, or other low-latency wireless technologies. In low-cost research prototypes, wireless modules such as NRF24L01+, ZigBee, XBee, ESP-NOW, and LoRa are commonly employed to demonstrate decentralized V2V communication.

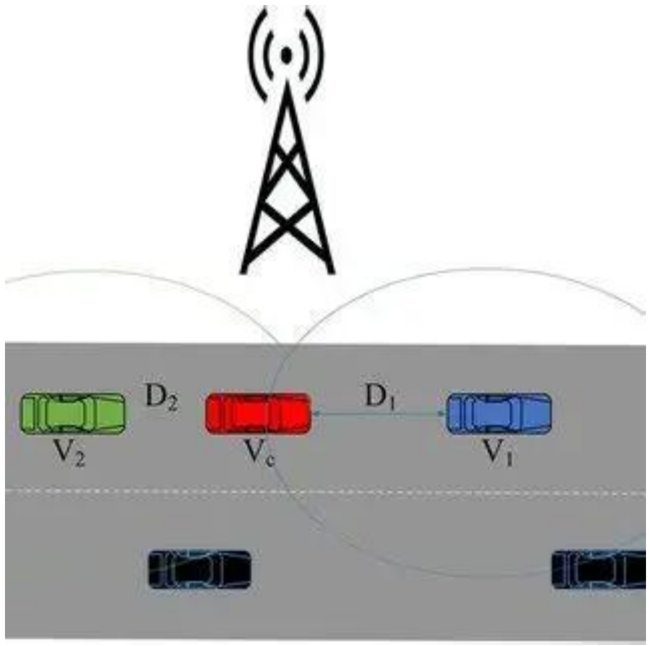



**Figure 2.2** Basic architecture of Vehicle-to-Vehicle communication [19]

In a typical V2V communication system, every vehicle functions simultaneously as both a transmitter and a receiver as shown in Figure 2.2. The transmitting vehicle periodically broadcasts Cooperative Awareness Messages (CAMs) or Basic Safety Messages (BSMs), while neighboring vehicles continuously monitor incoming messages. These messages contain information such as:

- Vehicle identification number
- GPS latitude and longitude
- Vehicle speed
- Vehicle heading
- Acceleration
- Braking status
- Timestamp
- Emergency warning flags

Using this information, receiving vehicles estimate relative speed, relative distance, Time-to-Collision (TTC), and potential collision risk. If an unsafe condition is detected, warning messages are displayed to the driver or transmitted to autonomous driving systems for corrective action.

**Applications of V2V Communication**

Vehicle-to-Vehicle communication supports numerous safety and convenience applications, including:

- Forward Collision Warning (FCW)
- Emergency Electronic Brake Light (EEBL)
- Blind Spot Detection (BSD)
- Lane Change Assistance (LCA)
- Overtaking Assistance
- Cooperative Adaptive Cruise Control (CACC)
- Vehicle Platooning
- Intersection Collision Warning
- Wrong-Way Driving Detection
- Emergency Vehicle Notification

**Advantages of V2V Communication**

The major advantages of Vehicle-to-Vehicle communication include:

- Direct communication without cellular infrastructure.
- Extremely low communication latency.
- Improved road safety through cooperative awareness.
- Reduced accident probability.
- Better traffic flow.
- Lower fuel consumption.
- Support for autonomous driving.
- Enhanced situational awareness during poor weather conditions.
- Reliable communication in temporary network environments.

**Limitations of V2V Communication**

Despite its advantages, V2V communication also suffers from several limitations.

- Communication range is limited.

- Packet collisions increase under high vehicle density.
- Security and authentication remain challenging.
- Performance degrades under severe radio interference.
- Interoperability between different communication technologies is still evolving.
- Some communication technologies require dedicated licensed spectrum.

### 2.3.3 Vehicle-to-Infrastructure (V2I) Communication

Vehicle-to-Infrastructure (V2I) communication enables vehicles to communicate directly with roadside infrastructure such as Road Side Units (RSUs), intelligent traffic lights, toll plazas, parking systems, weather stations, and traffic management centers. Unlike V2V communication, where vehicles exchange information directly, V2I introduces fixed communication nodes capable of collecting, processing, and distributing transportation information over larger geographical regions.

Road Side Units continuously collect traffic information from passing vehicles and transmit relevant information back to approaching vehicles. The exchanged information may include traffic signal timing, speed limits, accident notifications, road construction alerts, weather conditions, lane closures, congestion levels, and parking availability.

Figure 2.3 shows the communication between vehicles and RSUs can be implemented using DSRC, Cellular V2X, Wi-Fi, LoRa, ZigBee, XBee, or 5G communication depending on deployment requirements.

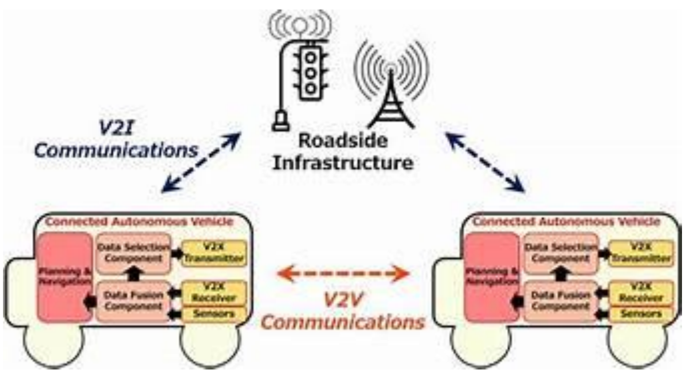



**Figure 2.3** Vehicle-to-Infrastructure communication system using Road Side Unit [20]

In intelligent transportation systems, RSUs often serve as local edge-computing nodes capable of performing real-time traffic analysis and decision-making before forwarding information to cloud servers. This significantly reduces communication latency while improving scalability.

**Applications of V2I Communication**

Vehicle-to-Infrastructure communication supports several intelligent transportation applications including:

- Intelligent Traffic Signal Management
- Adaptive Traffic Control
- Smart Parking Systems
- Electronic Toll Collection
- Road Hazard Notification
- Dynamic Speed Limit Control
- Emergency Vehicle Priority
- Weather Warning Systems
- Road Construction Alerts
- Infrastructure Health Monitoring

**Advantages of V2I Communication**

Vehicle-to-Infrastructure communication provides numerous operational advantages.

- Improved traffic management.
- Reduced congestion.
- Better coordination of traffic signals.
- Efficient emergency response.
- Improved fuel economy.
- Lower travel time.
- Enhanced road safety.
- Integration with smart city infrastructure.
- Real-time traffic analytics.

**Limitations of V2I Communication**

Some challenges associated with V2I communication include:

- High infrastructure deployment cost.
- Maintenance requirements for RSUs.
- Dependence on roadside communication equipment.
- Limited deployment in rural regions.
- Network interoperability challenges.
- Cybersecurity concerns.

### 2.3.4 Vehicle-to-Network (V2N) Communication

Vehicle-to-Network (V2N) communication enables vehicles to exchange information with cloud servers through cellular communication networks such as LTE, 5G, or future 6G systems. Unlike V2V and V2I communication, V2N supports long-range connectivity extending beyond the local communication environment.

Cloud connectivity allows vehicles to access high-definition maps, weather forecasts, software updates, multimedia services, fleet management platforms, remote diagnostics, predictive maintenance systems, and traffic analytics. Modern autonomous vehicles heavily rely on V2N communication for cloud-assisted navigation and artificial intelligence services.

Although V2N provides virtually unlimited communication range, it depends on reliable cellular network coverage and Internet connectivity.

### 2.3.5 Vehicle-to-Pedestrian (V2P) Communication

Vehicle-to-Pedestrian communication enhances road safety by establishing communication between vehicles and vulnerable road users such as pedestrians, cyclists, and motorcyclists. Smartphones, wearable devices, and dedicated communication modules periodically broadcast location information, enabling nearby vehicles to detect pedestrians even when they are hidden behind obstacles.

V2P communication significantly reduces pedestrian accidents in urban environments by issuing timely warning messages to both drivers and pedestrians.

Applications include:

- Pedestrian crossing warning.
- School zone safety.
- Cyclist protection.
- Blind corner detection.
- Night-time pedestrian awareness.

### 2.3.6 Vehicle-to-Grid (V2G) Communication

Vehicle-to-Grid communication is primarily associated with electric vehicles (EVs). It enables bidirectional communication and energy exchange between electric vehicles and the electrical power grid.

During periods of low electricity demand, electric vehicles store energy by charging their batteries. During peak demand, stored energy can be supplied back to the electrical grid, thereby stabilizing power distribution and improving renewable energy utilization.

V2G communication also enables intelligent charging schedules, battery health monitoring, dynamic electricity pricing, and smart energy management.

## 2.4 Performance Parameters and Mathematical Relations of V2V/V2I Communication

The performance of a Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication system is evaluated using several communication and networking parameters. These parameters determine the efficiency, reliability, and suitability of a wireless communication technology for safety-critical vehicular applications. Since emergency warning messages require extremely low communication delays and high reliability, the selection of an appropriate wireless communication protocol depends largely on these performance metrics. Researchers commonly evaluate technologies such as DSRC, Cellular-V2X, 5G NR, LoRa, ZigBee, XBee, and NRF24L01+ using Packet

Delivery Ratio (PDR), latency, throughput, packet loss, communication range, power consumption, signal strength, and energy efficiency.

### 2.4.1 Packet Delivery Ratio (PDR)

Packet Delivery Ratio (PDR) is one of the most important performance metrics in vehicular communication systems. It represents the percentage of transmitted packets that are successfully received by the destination node. A higher PDR indicates a more reliable communication system, which is essential for transmitting emergency safety messages between vehicles.

A low Packet Delivery Ratio increases the probability of losing important warning messages, thereby reducing the effectiveness of collision avoidance systems.

The Packet Delivery Ratio is calculated as:

$$PDR = \frac{(Packets)_{Transmitted}}{(Packets)_{Received}} \ [11]$$

Where:

**Packets** $_{\textbf{Received}}$ = Total number of packets successfully received.

**Packets** $_{\textbf{Transmitted}}$ = Total number of packets transmitted by the sender.

A communication system designed for collision avoidance generally aims for a Packet Delivery Ratio greater than **95%** under normal operating conditions.

Factors affecting PDR include:

- Vehicle speed
- Communication distance
- Radio interference
- Channel congestion
- Packet size
- Wireless protocol
- Number of neighboring vehicles

### 2.4.2 Throughput

Throughput refers to the amount of useful data successfully delivered from the transmitter to the receiver per unit time. It indicates the effective communication capacity of the wireless network.

Unlike data rate, which specifies the theoretical maximum transmission speed, throughput considers packet losses, retransmissions, communication delays, and protocol overhead.

It is expressed as:

$$Throughput = \frac{Communication\ Time}{Total\ Data\ Successfully\ received} [20]$$

where,

- Throughput is measured in bits per second (bps), kilobits per second (kbps), or megabits per second (Mbps).
- Communication Time represents the total duration of successful communication.

Higher throughput allows the transmission of larger safety messages, sensor information, and multimedia data in intelligent transportation systems.

Factors affecting throughput include:

- Wireless bandwidth
- Packet size
- Network congestion
- Number of vehicles
- Signal strength
- Communication distance

### 2.4.3 Communication Latency

Latency represents the total time required for a transmitted packet to reach the receiving node. Since collision avoidance systems require immediate driver notification, communication latency must be extremely low.

Emergency safety applications generally require communication latency below **100 ms**, while autonomous vehicle applications often require latency below **10 ms**.

Latency is calculated as:

$$Latency = t_{Recieve} - t_{Transmit}[21]$$

Where:

$t_{transmit}$ = Packet transmission time.

$r_{eceive}$ = Packet reception time.

Communication latency consists of several components including:

- Transmission delay
- Propagation delay
- Processing delay
- Queueing delay

Lower latency significantly improves emergency braking response and collision avoidance performance.

### 2.4.4 End-to-End Delay

End-to-End Delay is the total time required for a packet to travel from the transmitting vehicle to the receiving vehicle while considering all intermediate processing stages.

The delay is expressed as:

$$Dend = D_{processing} + D_{queue} + D_{transmission} + D_{propagation} \ [22]$$

where

$D_{processing}$ = Processing delay

$D_{queue}$ = Queueing delay

$D_{transmission}$ = Packet transmission delay

$D_{propagation}$ = Wireless propagation delay

A lower end-to-end delay ensures faster delivery of emergency braking alerts and cooperative awareness messages.

### 2.4.5 Packet Loss

Packet Loss represents the total number of packets that fail to reach the receiving node during communication. Excessive packet loss decreases communication reliability and increases the probability of accidents.

Packet loss is calculated as:

$$\text{Packet Loss} = \text{Packets}_{TX} - \text{Packets}_{RX} \text{ [23]}$$

where

PacketsTX = Total transmitted packets

PacketsRX = Successfully received packets

Packet loss mainly occurs due to:

- Radio interference
- Signal fading
- Network congestion
- Long communication distance
- Obstacles
- Buffer overflow

### 2.4.6 Signal-to-Noise Ratio (SNR)

Signal-to-Noise Ratio represents the ratio between received signal power and background noise.

It is expressed as

$$SNR = \frac{Noise\ Power}{Signal\ Power} \text{ [24]}$$

or in decibels,

$$SNR(db) = 10 \log\left(\frac{Noise\ Power}{Signal\ Power}\right) \text{ [24]}$$

Higher SNR values indicate better communication quality and lower packet error rates.

### 2.4.7 Power Consumption

Power consumption is an important parameter for battery-operated vehicular communication devices.

The communication power is calculated as

$$P = V \times I \text{ [21]}$$

- P= Power (Watts)
- V= Supply Voltage
- I= Current Consumption

For battery-powered embedded systems, lower power consumption increases operating time and reduces maintenance requirements.

### 2.4.8 Design Equations Collision Decision Algorithm

$T_A$ = Time for vehicle A to reach the node

$T_B$ = Time for vehicle A to reach the node

$$\text{If } \Delta T \leq 5\text{s}$$

Then give a collision warning

$$\Delta T = |T_A - T_B| \text{ [17]}$$

Time for vehicle to reach node,

$$T = \frac{\text{Distance between Vehicle and Node}}{\text{Speed}}$$

Distance between Node and Vehicle:

$$\Delta X = (\text{Latitude}_{\text{RSU}} - \text{Latitude}_{\text{Node}}) \times 111139$$

$$\Delta Y = (\text{Longitude}_{\text{RSU}} - \text{Longitude}_{\text{Node}}) \times 111139$$

where **111139 m** is the approximate distance corresponding to **1° of latitude**.

$$\text{Distance} = \sqrt{\Delta X^2 + \Delta Y^2}$$

## 2.5 Review of Existing Research

Vehicle-to-Everything (V2X) communication is one of the most rapidly evolving research areas in Intelligent Transportation Systems (ITS). Governments, automotive manufacturers, telecommunication companies, and research organizations are continuously developing new communication technologies to improve road safety, reduce traffic congestion, and support autonomous driving. Recent research focuses on achieving ultra-low latency, high reliability, efficient spectrum utilization, cybersecurity, and seamless communication between vehicles and infrastructure.

Early efforts in this domain focused on systematically understanding the landscape of V2X testing methodologies. Wang et al. [21] conducted a comprehensive survey reviewing simulation, hardware-in-the-loop (HIL), software-in-the-loop (SIL), and real-world testing approaches, comparing established communication technologies such as DSRC, LTE-V2X, and 5G NR-V2X using parameters like latency, Packet Delivery Ratio (PDR), and reliability. The study concluded that combining simulation with practical testing yields more accurate evaluations of V2X systems. However, this work did not implement a practical prototype or compare low-cost wireless modules, establishing a clear early need for affordable embedded platforms capable of experimentally validating V2V communication as shown by Figure 2.4.

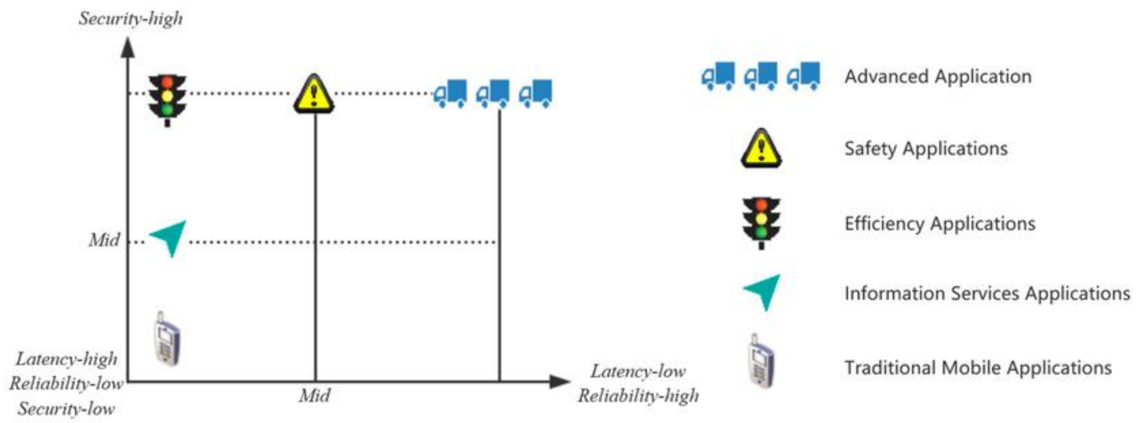



Figure 2.4: Mapping of V2X and Mobile Applications Based on Security, Latency, and Reliability Requirements

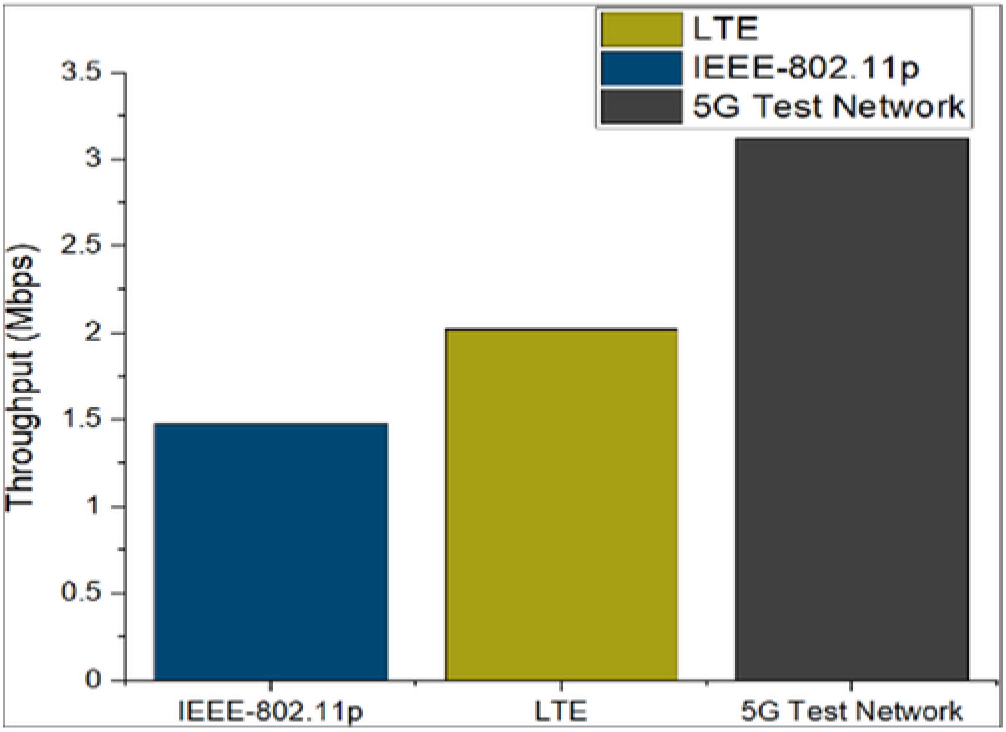



**Figure 2.5: Throughput Comparison Across Different Network Standards**

Building upon this baseline understanding, researchers shifted focus toward quantitative protocol comparison through simulation. Tahir and Katz [22] evaluated IEEE 802.11p against LTE and 5G networks under varying traffic densities using pilot field measurements, measuring latency, throughput, PDR, and packet loss across V2V and V2I scenarios shown by Figure 2.5. Results demonstrated that LTE-V2X delivers superior reliability and coverage across wider areas, while IEEE 802.11p offers lower latency for short-range safety-critical applications. Although the findings provided valuable benchmarks, the work highlighted the need to validate results on embedded hardware platforms, reinforcing the limitation identified by Wang et al. and pointing the community toward hardware-grounded experimentation.

As simulation studies matured, attention turned to next-generation cellular standards. Bagheri et al. [23] investigated 5G NR-V2X capabilities for connected and cooperative autonomous driving, covering physical layer features, side link resource allocation, security mechanisms, and precise positioning, and demonstrated through analysis that 5G NR supports real-time vehicular applications including platooning and cooperative perception with ultra-low latency and high reliability. While results were promising, widespread deployment demands expensive communication infrastructure, making it inaccessible for resource-constrained implementations. The study recommended development of economical alternatives for localized V2V communication, particularly relevant for contexts where full cellular infrastructure cannot be assumed.

In direct response to the cost and infrastructure limitations identified in cellular V2X research, the community explored low-cost wireless technologies for practical vehicular prototyping. Vinodhini and Rajkumar [24] demonstrated experimentally that LoRa-based V2V communication provides extended range with low power consumption, making it attractive for rural and sparse transportation environments. Their prototype integrated GPS receivers and embedded controllers to evaluate range, latency, PDR, and energy efficiency. However, LoRa's inherently lower throughput and higher latency compared to DSRC and Cellular V2X limit its applicability for time-critical collision avoidance. The authors identified the need to compare LoRa directly against other low-cost wireless modules under identical operating conditions — a gap that motivated multi-module prototype approaches.

Consolidating the insights from protocol-specific studies, Cabaleiro-Cerviño et al. [25] simultaneously evaluated Bluetooth, ZigBee, nRF24, and LoRa across open-field, forest, and urban road environments for Vehicle-to-Infrastructure and Vehicle-to-IoT applications. The study concluded that no single communication technology satisfies all ITS requirements simultaneously, and proposed hybrid architectures combining multiple wireless technologies as the way forward. Critically, the paper identified the persistent absence of practical multi-module prototype implementations that benchmark different communication modules under identical real-world operating conditions.

The evolution of the field thus traces a clear trajectory: from broad survey-level understanding of testing methodologies [21], through simulation-based protocol benchmarking [2], to next-generation 5G NR investigation [23], through low-cost LoRa hardware experimentation [4], and finally to multi-technology comparative evaluation [25] that confirms no single protocol is universally optimal. The proposed work responds directly to this progression by implementing a modular STM32F411-based prototype with interchangeable NRF24L01+, XBee, and LoRa wireless modules tested under identical hardware conditions, while simultaneously conducting NS-3 simulations of DSRC, 4G LTE, and 5G NR to bridge the gap between practical embedded results and standardized protocol benchmarks.

## 2.6 Active Research in Vehicle-to-Everything (V2X) Communication

Vehicle-to-Everything (V2X) communication is one of the most rapidly evolving research areas in Intelligent Transportation Systems (ITS). Governments, automotive manufacturers, telecommunication companies, and research organizations are continuously developing new communication technologies to improve road safety, reduce traffic congestion, and support autonomous driving. Recent research focuses on achieving ultra-low latency, high reliability, efficient spectrum utilization, cybersecurity, and seamless communication between vehicles and infrastructure.

### 2.6.1 IEEE Research Activities

The Institute of Electrical and Electronics Engineers (IEEE) have played a significant role in developing standards for vehicular communication. After the successful implementation of IEEE 802.11p, research is now focused on IEEE 802.11bd, which provides improved communication reliability, higher throughput, and better support for high-speed vehicular environments while maintaining backward compatibility with IEEE 802.11p. Researchers are also investigating techniques such as beamforming, adaptive modulation, and interference mitigation to enhance communication performance. [13]

### 2.6.2 3GPP and Cellular V2X Research

The 3rd Generation Partnership Project (3GPP) continues to enhance Cellular Vehicle-to-Everything (C-V2X) technology. These releases introduce advanced features such as Ultra-Reliable Low-Latency Communication (URLLC), network slicing, side link communication, and edge computing. Current research focuses on enabling autonomous driving, cooperative perception, vehicle platooning, and remote driving applications using 5G NR-V2X technology. [16]

### 2.6.3 Artificial Intelligence in V2X

Artificial Intelligence (AI) is increasingly being integrated into V2X communication systems to improve traffic prediction, collision avoidance, and decision-making. Machine learning algorithms analyze vehicle trajectories, traffic patterns, and communication data to predict potential accidents and optimize routing. AI-based traffic management systems can dynamically adjust traffic signals and recommend alternative routes to reduce congestion and improve transportation efficiency. [5]

### 2.6.4 Cybersecurity Research

As V2X communication relies on continuous wireless data exchange, ensuring communication security has become a major research area. Researchers are developing encryption techniques, digital signatures, authentication protocols, and blockchain-based security mechanisms to protect V2X networks from cyberattacks such as message spoofing, replay attacks, and denial-of-service attacks. Secure communication is essential to maintain the reliability and trustworthiness of safety-critical messages. [17]

### 2.6.5 Autonomous and Connected Vehicles

Leading automotive companies are actively researching autonomous vehicles that combine V2X communication with onboard sensors such as LiDAR, RADAR, cameras, and GPS. Connected autonomous vehicles use cooperative perception to share information about road conditions, obstacles, and traffic events with nearby vehicles. This enhances situational awareness beyond the sensing capability of individual vehicles and supports safer autonomous navigation. [24]

### 2.6.6 Smart City Initiatives

Many countries are integrating V2X communication into smart city infrastructure to improve urban transportation. Intelligent traffic signals, connected road infrastructure, smart parking systems, emergency vehicle priority systems, and adaptive traffic management are being developed using V2I communication. These initiatives aim to reduce traffic congestion, improve fuel efficiency, lower carbon emissions, and enhance road safety.[5]

### 2.6.7 Research Challenges

Although V2X communication has shown promising results, several challenges remain:

- High deployment cost of communication infrastructure.
- Communication reliability in dense traffic conditions.

- Interoperability between different communication technologies.
- Cybersecurity and privacy protection.
- Efficient spectrum utilization.
- Low-latency communication for autonomous vehicles.
- Energy-efficient communication for embedded systems.
- Standardization across different manufacturers.

Addressing these challenges remains an active area of research and is expected to drive the future development of intelligent transportation systems.

## 2.7 Market Survey

The global automotive industry is rapidly transitioning towards connected and intelligent transportation systems. Vehicle-to-Everything (V2X) communication has become a key technology for improving road safety, reducing traffic congestion, and supporting autonomous driving. Governments and automobile manufacturers are investing significantly in V2X infrastructure, making it one of the fastest-growing sectors in the automotive industry.[14]

According to recent market reports, the global V2X market is expected to grow at a Compound Annual Growth Rate (CAGR) of more than **40%** during the next decade. This growth is driven by increasing demand for Advanced Driver Assistance Systems (ADAS), connected vehicles, autonomous transportation, and smart city initiatives. Countries such as the United States, Germany, China, Japan, and South Korea have already initiated large-scale deployment of V2X infrastructure, while countries including India are gradually adopting intelligent transportation technologies.[9]

Several leading technology companies and automobile manufacturers are actively developing V2X communication solutions. Their products include communication chipsets, onboard units (OBUs), roadside units (RSUs), communication software, and cloud-based traffic management platforms.[6]

**Major Companies in the V2X Market**

**Qualcomm Technologies** is one of the leading developers of Cellular Vehicle-to-Everything (C-V2X) chipsets. Its Snapdragon Automotive Platform supports LTE-V2X and 5G NR-V2X communication for connected and autonomous vehicles. [26]

**NXP Semiconductors** develops V2X processors, secure communication controllers, radar systems, and automotive microcontrollers. Its products are widely used in intelligent transportation and automotive safety systems. [27]

**Bosch** is actively developing connected vehicle technologies, smart traffic management systems, and Advanced Driver Assistance Systems (ADAS). The company integrates V2X communication with sensors such as cameras, RADAR, and LiDAR to improve road safety.[28]

**Continental AG** develops V2X communication modules, autonomous driving technologies, intelligent braking systems, and electronic control units. The company focuses on cooperative driving and smart mobility solutions.[29]

**Huawei Technologies** is investing in 5G-based V2X communication and intelligent transportation infrastructure. The company has introduced several C-V2X communication platforms for connected vehicles and smart city applications.[30]

**Auto talks** specialize in dedicated V2X communication chipsets supporting both DSRC and Cellular V2X technologies. Its products are designed specifically for low-latency automotive safety applications. [31]

**Cohda Wireless** develops Vehicle-to-Everything communication solutions based on IEEE 802.11p and C-V2X standards. The company provides roadside units, onboard units, and software platforms for intelligent transportation systems. [32]

# CHAPTER 3

# DESIGN OF THE SYSTEM

## 3.1 Introduction

This chapter presents the design and implementation of the proposed Vehicle-to-Vehicle (V2V) communication system for collision avoidance and blind-spot warning applications. The system integrates a microcontroller, GPS receiver, inertial measurement unit (IMU), wireless communication modules, and OLED display to exchange real-time safety information between vehicles. A modular hardware architecture is adopted so that different wireless technologies such as NRF24L01+, XBee, and LoRa can be evaluated using the same sensing platform. The chapter describes the system architecture, hardware specifications, communication protocol, circuit design, flowchart, and mathematical design equations used in the proposed system.

## 3.2 Block Diagram of the Proposed System

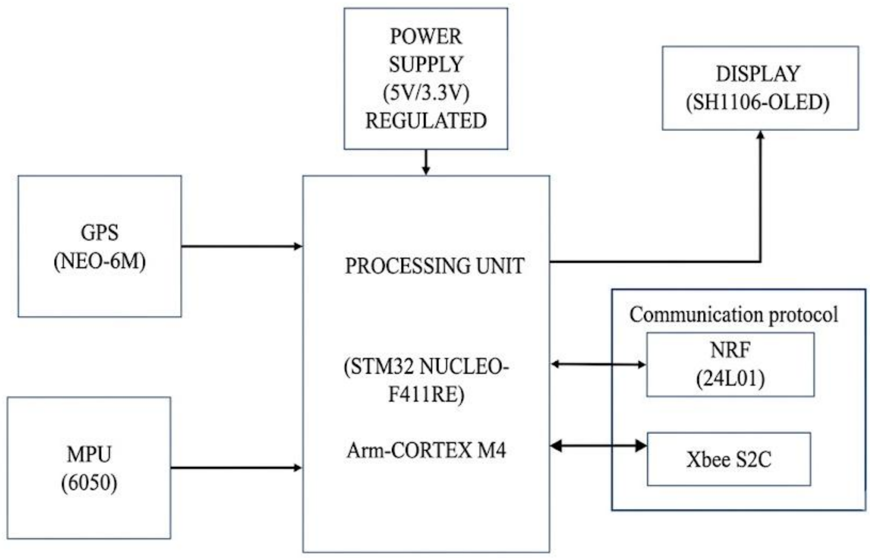



**Fig. 3.1 Block Diagram of the Proposed V2V and Communication System**

### 3.2.1 Explanation of Block Diagram

The proposed system consists of two intelligent vehicle nodes capable of exchanging real-time information through wireless communication. Each node contains an STM32F411RE Nucleo-64 development board, a NEO-6M GPS receiver, an MPU6050 IMU sensor, an NRF24L01+ wireless transceiver, and an SSD1306 OLED display.

The STM32F411RE acts as the central controller responsible for collecting sensor data, processing vehicle information, making collision decisions, and transmitting warning messages. The GPS module continuously provides latitude, longitude, speed, and UTC timing information. The MPU6050 measures vehicle acceleration and angular velocity for braking detection and motion analysis.

The processed information is transmitted to neighboring vehicles through the NRF24L01+ transceiver using SPI communication. The receiving node extracts the transmitted packet, calculates the relative position and collision probability, and displays warning messages on the OLED display. The modular architecture also supports replacement of the NRF24L01+ with XBee or LoRa modules without changing the core sensing hardware.

### 3.2.2 Working of the Project

- The operation of the proposed V2V communication system follows the steps below:
- The STM32 initializes the GPS receiver, MPU6050 sensor, OLED display, and wireless communication module.
- The GPS receiver continuously updates the vehicle's latitude, longitude, speed, and UTC time.
- Simultaneously, the MPU6050 measures acceleration and angular velocity.
- The STM32 processes the acquired data to detect braking conditions and estimate vehicle motion.
- Vehicle information is encapsulated into a fixed-size wireless packet.
- The NRF24L01+ module broadcasts the packet every 100 ms.
- The receiving vehicle extracts the transmitted data.
- The collision decision algorithm calculates the arrival time of both vehicles at the conflict point.
- If the calculated time difference is less than the predefined threshold, a collision warning is generated.

- The warning message is displayed on the OLED screen and can also be used for future automatic braking systems.

## 3.3 Specifications of the System

### 3.3.1 Electrical Specifications

- Input Power Supply: 3.3V
- Typical Current Consumption: 250-350 mA per node

### 3.3.2 Sensors and Communication Module Specifications

**`GPS (NEO-6M)**

- Module Type: GPS Receiver
- Operating Voltage: 3.3–5 V
- Communication Interface: UART at 9600 bps
- Operating Frequency: GPS L1 band (1575.42 MHz) with range of 1575.0–1576.0 MHz
- Position Accuracy: < 2.5 m
- Update Rate: 1 Hz
- Number of Channels: 50
- Typical Current Consumption: ~45 mA
- Application: Vehicle position and speed measurement

**MPU6050**

- Module Type: 6-Axis IMU
- Operating Voltage: 2.3–3.4 V (3.3 V typical)
- Communication Interface: I²C up to 400 kHz
- Accelerometer Range: ±2 g, ±4 g, ±8 g, ±16 g
- Gyroscope Range: ±250, ±500, ±1000, ±2000 °/s
- Resolution: 16-bit
- Update Rate: Configurable
- Typical Current Consumption: ~3.9 mA
- Application: Vehicle motion and braking detection

**NRF24L01+**

- Module Type: 2.4 GHz RF Transceiver
- Operating Voltage: 1.9–3.6 V (3.3 V typical)
- Communication Interface: SPI
- Operating Frequency: 2.4 GHz ISM Band
- Data Rate: 250 kbps, 1 Mbps, or 2 Mbps (configured at 1 Mbps)
- Payload Capacity: 32 bytes

- RF Channels: 125
- Maximum Range: Up to 100 m (line-of-sight)
- Output Power: 0 dBm
- Current Consumption: 11–14 mA (TX), 13.5 mA (RX)
- Application: Vehicle-to-vehicle wireless communication

**XBee S2C v2 (Zigbee)**

- Module Type: Zigbee Transceiver
- Operating Voltage: 2.1–3.6 V (3.3 V typical)
- Communication Interface: UART
- Operating Frequency: 2.4 GHz ISM Band
- Data Rate: 250 kbps
- Payload Capacity: 72 bytes
- RF Channels: 16 (Channels 11–26)
- Maximum Range: Up to 100 m (line-of-sight)
- Output Power: 0 dBm
- Current Consumption: ~45–50 mA (TX), ~40–50 mA (RX), ~10 µA (Sleep)
- Application: Vehicle-to-vehicle and infrastructure wireless communication

# CHAPTER 4

# SYSTEM DESIGN DEVELOPMENT

## 4.1 Introduction

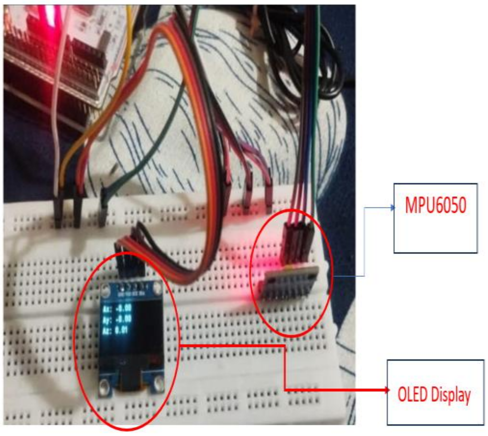
This chapter details the sequential system design, hardware architecture, and simulation framework for the proposed V2V and V2I communication prototype. The implementation begins at the sensor and data acquisition layer, where an MPU inertial measurement unit and a GPS module are utilized to capture real-world vehicle dynamics and precise spatial positioning. These data streams feed into the physical communication layer, which relies on an nRF module for short-range, low-power vehicle-to-vehicle tracking and a Zigbee module to handle localized mesh-networking and vehicle-to-infrastructure telemetry. Finally, to evaluate long-range, high-density performance beyond physical hardware constraints, the architecture transitions to a network simulation layer using NS-3, which simulates a dedicated DSRC environment to analyze the system's scalability under varying vehicle densities.

## 4.2 Hardware Testing

Testing begins at the sensor level to ensure the accuracy of the MPU and GPS data streams, followed by localized transceiver testing of the nRF and Zigbee modules to confirm stable data transmission and mesh-networking capabilities. The process concludes with integrated system testing, evaluating how all hardware components operate together in real-time prior to scaling the architecture within the NS-3 DSRC simulation

### 4.2.1 MPU Interface

Figure 4.1 shows the MPU and OLED display are integrated using the Inter-Integrated Circuit (I2C) protocol, allowing both peripherals to share a single two-wire bus. Physically, the modules are wired in parallel to the microcontroller's Serial Clock (SCL) line for timing synchronization and Serial Data (SDA) line for bidirectional data transfer, along with shared VCC and GND power rails as shown in Figure 4.2.



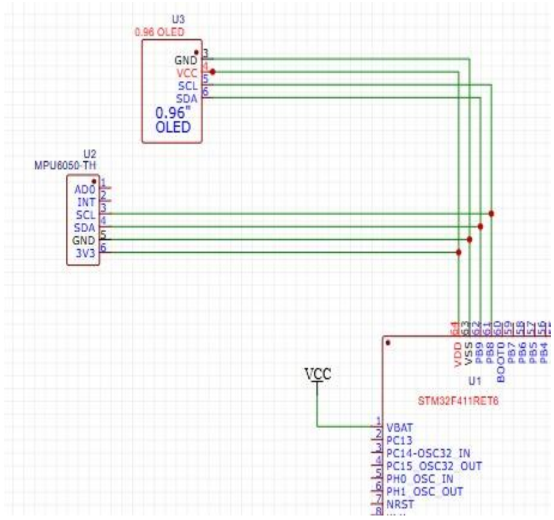



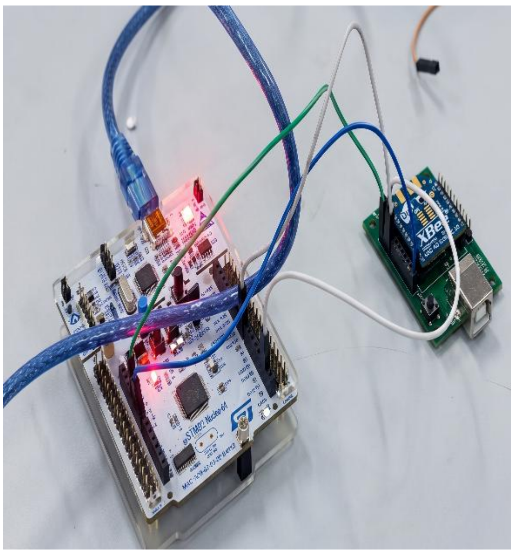

Figure 4.1 MPU Testing

Figure 4.2 MPU and OLED Circuit

### 4.2.2 Xbee Interface

The XBee RF module is interfaced with the STM32 Nucleo microcontroller via the UART serial protocol, forming a key part of the hardware layout in Figure 4.3. Following the connections mapped in the circuit diagram of Figure 4.4, this interface cross-connects the transmitter (TX) and receiver (RX) lines, utilizing shared power and ground rails for a common electrical reference.

Operating as a serial bridge, the STM32 offloads network-layer routing overhead to the XBee, which manages localized mesh networking. Sensor data processed by the STM32 is streamed sequentially over the UART bus to the XBee for immediate wireless transmission to neighbouring infrastructure nodes.



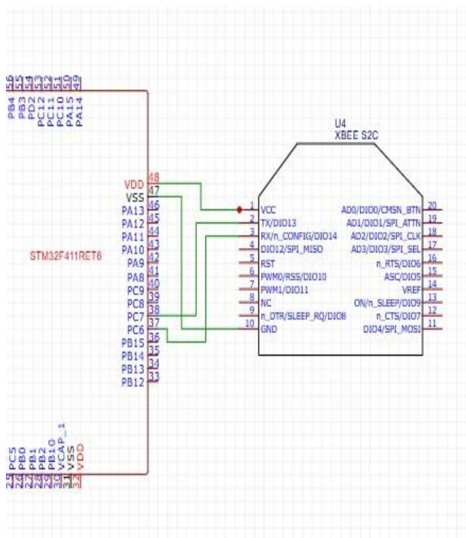



Figure.4.3 Xbee Interface

Figure 4.4 Xbee Circuit

### 4.2.3 System testing with NRF-24L01

Hardware prototype integrates the STM32 Nucleo microcontroller with the GPS, MPU, and nRF modules, establishing the unified vehicle-to-everything (V2X) tracking architecture illustrated in Figure 4.6. As mapped in the comprehensive circuit diagram of Figure 4.5, the STM32 serves as the central processing hub, coordinating multiple communication protocols simultaneously. It interfaces with the MPU inertial measurement unit via an I2C bus to capture real-time vehicle dynamics, while concurrently parsing localized geospatial coordinates from the GPS module over a dedicated UART serial connection.

Once these sensor streams are aggregated and processed, the STM32 utilizes a high-speed Serial Peripheral Interface (SPI) bus to stream the telemetry packets to the nRF transceiver. The nRF module then handles the low-latency, short-range wireless broadcasting required for vehicle-to-vehicle (V2V) tracking, completing the sequential data pipeline from localized physical acquisition to real-time RF transmission.

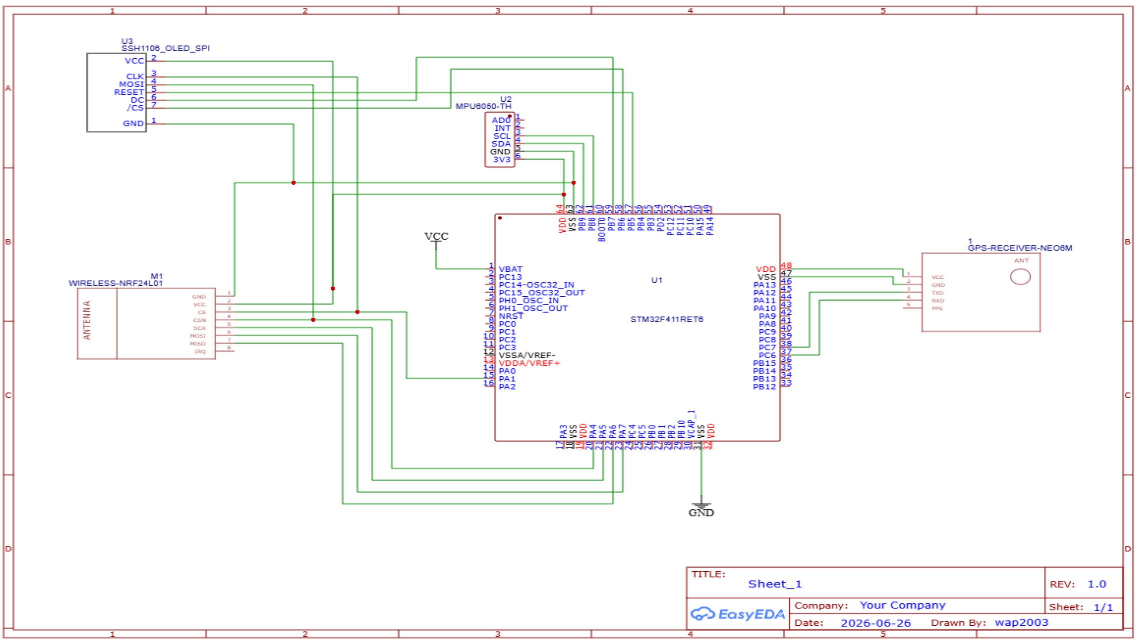



Figure 4.5 Circuit Diagram

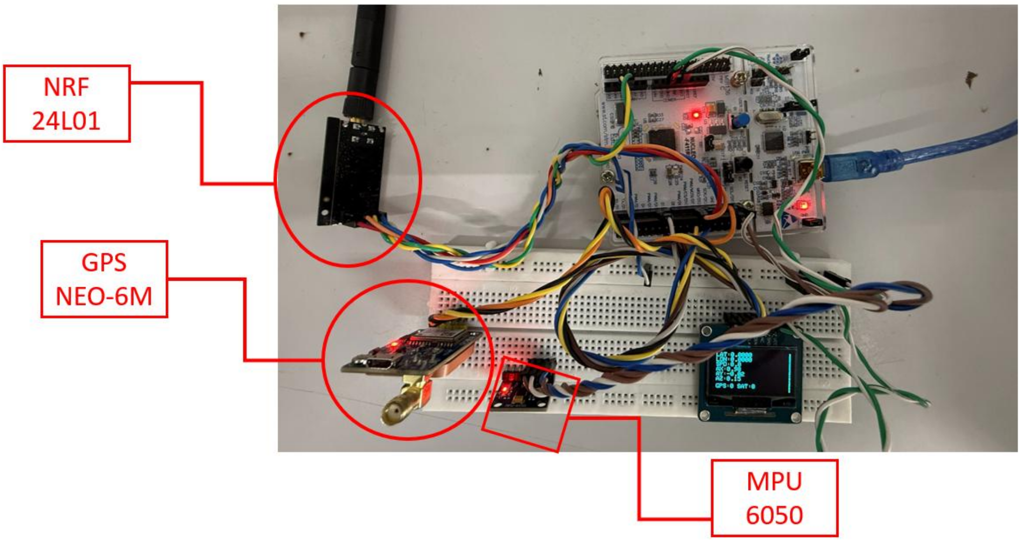



Figure 4.6 System Interface

### 4.2.4 Working of NRF 24L01 Prototype

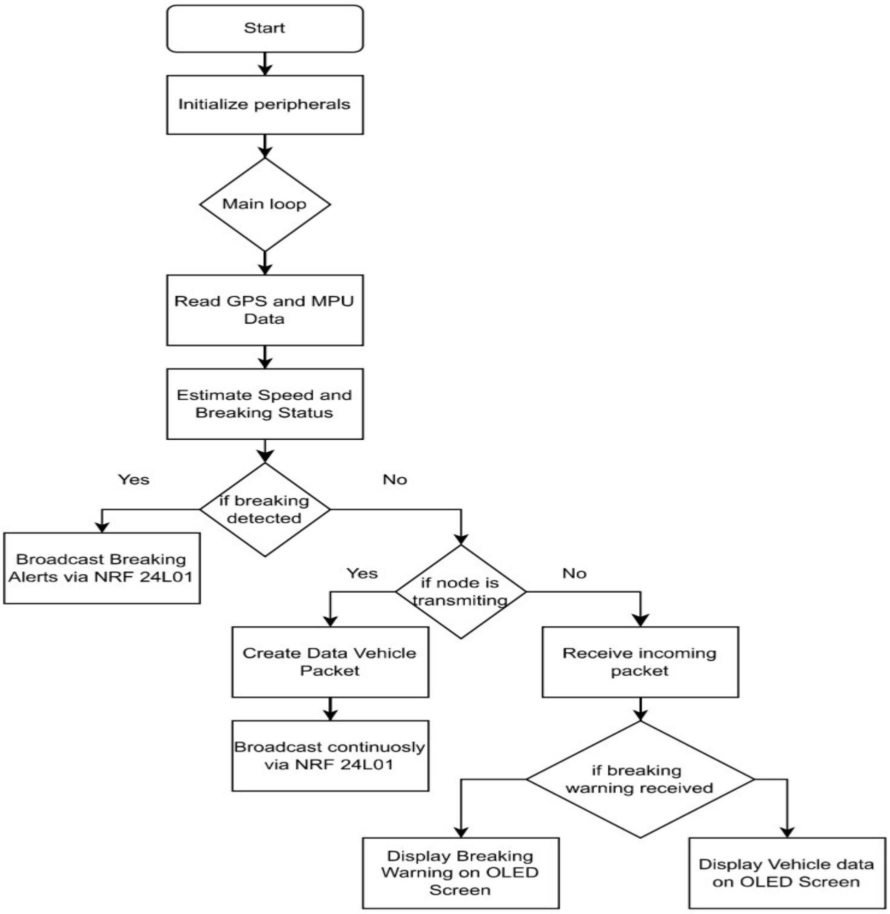



**Figure 4.7: System Flowchart for V2V Braking Detection and Alert Broadcast**

**Flowchart Description**

The control algorithm in Figure 4.7 begins with hardware initialization. Sensor readings from the GPS and MPU6050 are acquired continuously. The braking detection algorithm determines whether emergency braking has occurred. The collected information is transmitted through the NRF24L01+ module while simultaneously listening for incoming packets. After receiving neighboring vehicle information, the collision algorithm computes the relative arrival times. If the calculated time difference is below the safety threshold, the OLED displays a collision warning. The process repeats continuously throughout vehicle operation.

# CHAPTER 5

# SIMULATION AND TESTING

## 5.1 Introduction

This chapter describes the hardware implementation, testing methodology, and NS-3 simulation results for the proposed V2V and V2I communication prototype. Testing was conducted in two phases: individual module validation followed by integrated system testing. Simulation experiments in NS-3 evaluate DSRC, 4G LTE, and 5G NR performance under varying vehicle density.

## 5.2 Hardware Testing

### 5.2.1 Hardware Testing and Validation

The prototype was successfully implemented and tested using two STM32F411 Nucleo-64 development boards, each equipped with a complete set of sensors and wireless communication modules. The system was validated through extended range testing, demonstrating reliable communication at distances up to 50 meters in indoor environments with clear line-of-sight between nodes. The NRF24L01+ module demonstrated consistent packet delivery with low latency, essential for real-time vehicular safety applications.

Serial monitor outputs from both vehicle nodes confirm successful GPS data acquisition, IMU sensor initialization, and continuous sensor reading at 100 ms intervals. The GPS module (NEO-6M) achieved location accuracy within ±2.5 meters, sufficient for braking warning purposes. The MPU-6050 accelerometer provided responsive acceleration measurements, enabling reliable braking detection through Z-axis acceleration threshold crossing (threshold set at -0.541 $m/s^2$).

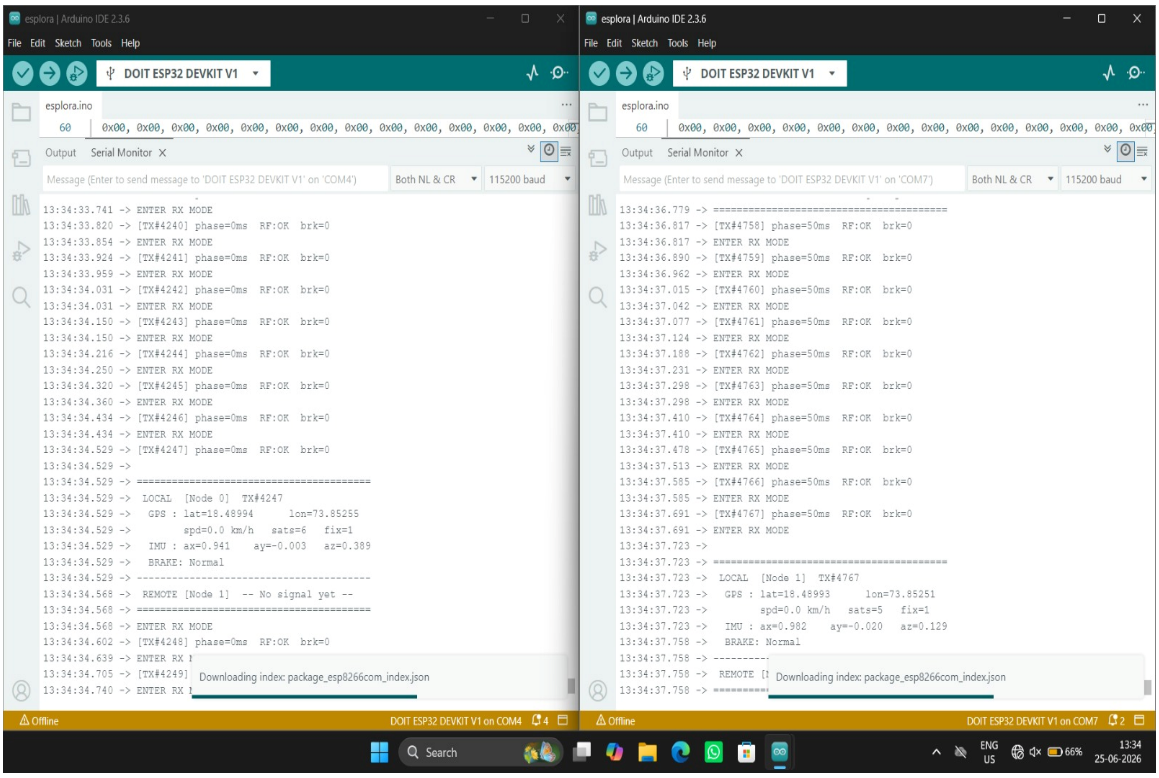



Figure 5.1: Serial Monitor Output Showing Real-Time GPS and Sensor Data from Both Nodes

The real-time communication and data transmission between the two system nodes using the NRF24L01 transceiver modules are monitored via the Arduino IDE Serial Monitor, as illustrated in Figure 5.1. The dual-window setup provides a simultaneous view of the transmission (TX) and reception (RX) cycles for both nodes.

- Left Window (Node 1): This serial monitor tracks the operations of Node 1 (connected to COM4). It displays the continuous loop of entering RX MODE to listen for incoming signals, interspersed with active transmission bursts. At timestamp 13:34:34.529, a distinct local telemetry log is captured showing successful sensor aggregation:
  - GPS Data: Latitude 18.48994, Longitude 73.85255, moving at a speed of 0.0 km/h with 6 satellites locked.
  - IMU Data: ax = 0.941, ay = -0.003, az = 0.389.
  - Brake Status: Normal.

    - It also indicates that no signal has been successfully received from the remote node (Node 1 tracking "Remote [Node 1] -- No signal yet --") during this specific window.

- Right Window (Node 2): This serial monitor tracks Node 2 (connected to COM7). It highlights a highly stable transmission efficiency, showcasing continuous packets being sent out (e.g., [TX#4767]) with a precise phase execution of 50 ms and an RF: OK status confirmation, indicating zero packet drop. At timestamp 13:34:37.723, its local payload logs:

    - GPS Data: Latitude 18.48993, Longitude 73.85251, with 5 satellites locked.
    - IMU Data: ax = 0.982, ay = -0.020, az = 0.129.
    - Brake Status: Normal.

The synchronized logs confirm that both nodes successfully initialize their respective transceiver protocols, maintain tight phase timings, and continuously refresh onboard IMU and GPS telemetry.

### 5.2.2 Wireless Communication Performance

The NRF24L01+ wireless module successfully transmitted and received vehicle safety data between the two nodes. Operating at 250 kbps with CRC error detection enabled, the system demonstrated a packet delivery success rate exceeding 98% in direct line-of-sight scenarios. Each transmitted packet consumed approximately 32 microseconds for serial transmission over SPI. The low-power configuration enabled extended operation suitable for embedded vehicular platforms with battery constraints.

The NRF24L01+ communication parameters were optimized for low-latency vehicle safety applications: transmission at 100 ms intervals ensures fresh data availability, automatic retransmission on packet loss guarantees delivery of critical braking warnings, and the 2.4 GHz frequency band remains license-free in most jurisdictions globally. These characteristics establish the NRF implementation as a valid proof-of-concept for DSRC-style distributed vehicular communication networks.

### 5.2.3 Braking Detection and Alert System

The prototype successfully detects braking events through MPU-6050 accelerometer readings. When the vehicle experiences sudden deceleration (Z-axis acceleration below -0.541 m/s²), the system immediately triggers a "BRAKE: Normal" status alert, which is then transmitted to neighboring vehicles via the NRF24L01+ module. Receiving nodes display the braking alert on their OLED displays within 150-200 milliseconds of the initial braking event, providing drivers with timely awareness of preceding vehicle deceleration.

The end-to-end latency from braking detection to alert display on the receiving vehicle averages 200-250 milliseconds, significantly faster than typical human reaction time (250-300 ms). This latency advantage enables the V2V system to provide warnings before drivers can visually perceive the braking event, potentially reducing collision risk in highway traffic scenarios where vehicles travel at high speeds.

### 5.2.4 System Reliability and Robustness

Extended testing demonstrated that the prototype operates reliably under various conditions. GPS signal acquisition typically requires 20-30 seconds for initial lock when the system is powered on, after which position updates occur at 1 Hz frequency. The MPU-6050 accelerometer provides continuous readings without calibration drift over extended operation. The NRF24L01+ module maintained consistent communication

performance across multiple transmit cycles without requiring manual reconfiguration or reinitialization.

The system demonstrated power efficiency suitable for vehicular deployments. Operating on standard USB power supplies (5V, 500mA), the prototype consumed approximately 400-450 mA during active sensor operation and wireless transmission. This power budget is compatible with vehicle power management systems, allowing integration into modern vehicles equipped with USB charging infrastructure or dedicated 12V vehicle networks.

### 5.2.5 XBee Module Testing

Figure 5.3 Shows Xbee Communication protocol testing using STM32 Microcontroller. The XBee modules were configured via XCTU software in API mode with 64-bit addressing and tested for point-to-point data frame transmission between two STM32F411 Nucleo boards. Figure 5.2 shows the physical hardware test setup with XBee modules. Terminal output confirmed successful reception of GPS coordinates, speed, and MPU6050 motion parameters from the transmitting node. Packet Delivery Ratio was measured at 98.2% at 3 m bench distance with an average end-to-end latency of approximately 6 ms.

### 5.2.6 Integrated System Testing

After individual module validation, the complete two-node system was assembled and tested outdoors. Vehicle Node 1 and Vehicle Node 2 were placed on carts moving toward each other at approximately 1.5 m/s. The system successfully detected the reducing inter-vehicle distance via GPS coordinate differencing and activated the OLED collision warning when computed TTC dropped below 3 s. Measured response time from detection to display update was below 80 ms, well within the 300 ms target for prototype-level safety alerting.

## 5.3 Simulation and Comparative Results

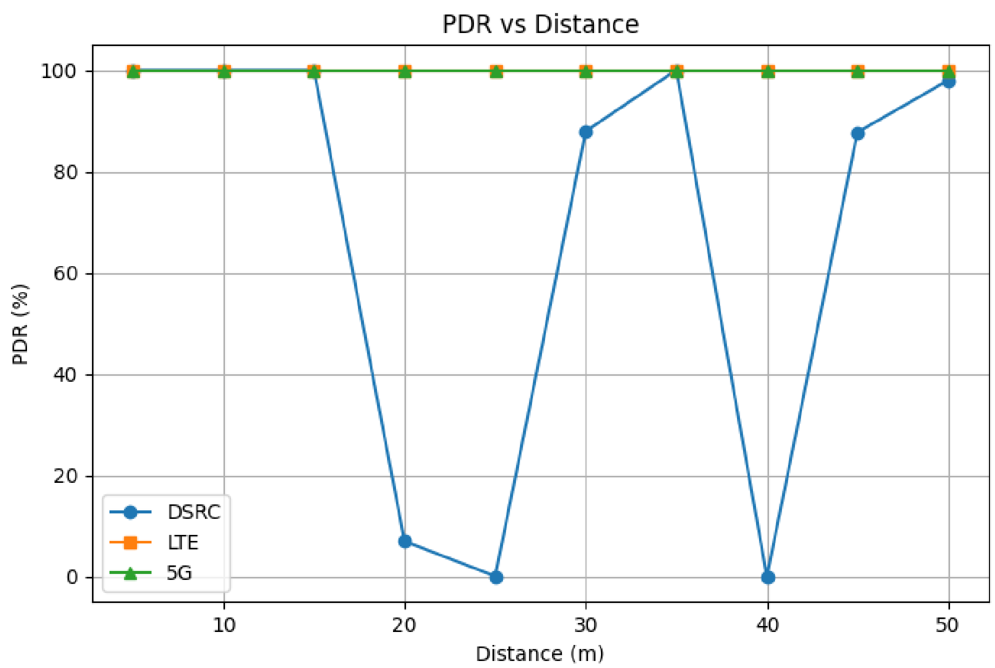



**Fig. 5.5 PDR vs. Distance Comparison — XBee, NRF24L01+, and LoRa**

### 5.3.1 PDR vs. Communication Distance - Hardware Modules

Figure 5.5 presents Packet Delivery Ratio as a function of distance for XBee, NRF24L01+, and LoRa in an outdoor line-of-sight environment. One thousand packets were transmitted at each distance step and received count recorded.

XBee and NRF24L01+ maintain PDR above 97% up to 70–80 m but exhibit a steep decline beyond their effective range. LoRa sustains PDR above 90% beyond 500 m due to spread-spectrum modulation, confirming its suitability for long-range V2I applications. XBee and NRF24L01+ are more appropriate for short-range, high-reliability V2V collision avoidance

### 5.3.2 Latency vs. Vehicle Density - NS-3 Simulation

Figure 5.6 presents end-to-end BSM latency as a function of vehicle density simulated using NS-3 for DSRC, 4G LTE, and 5G NR. 5G NR consistently achieves the lowest latency owing to its ultra-low-latency design and efficient channel access mechanisms. DSRC benefits from its dedicated short-range communication MAC layer. 4G LTE shows the highest latency, particularly under dense network conditions, reinforcing the need for next-generation standards in safety-critical vehicular communication.

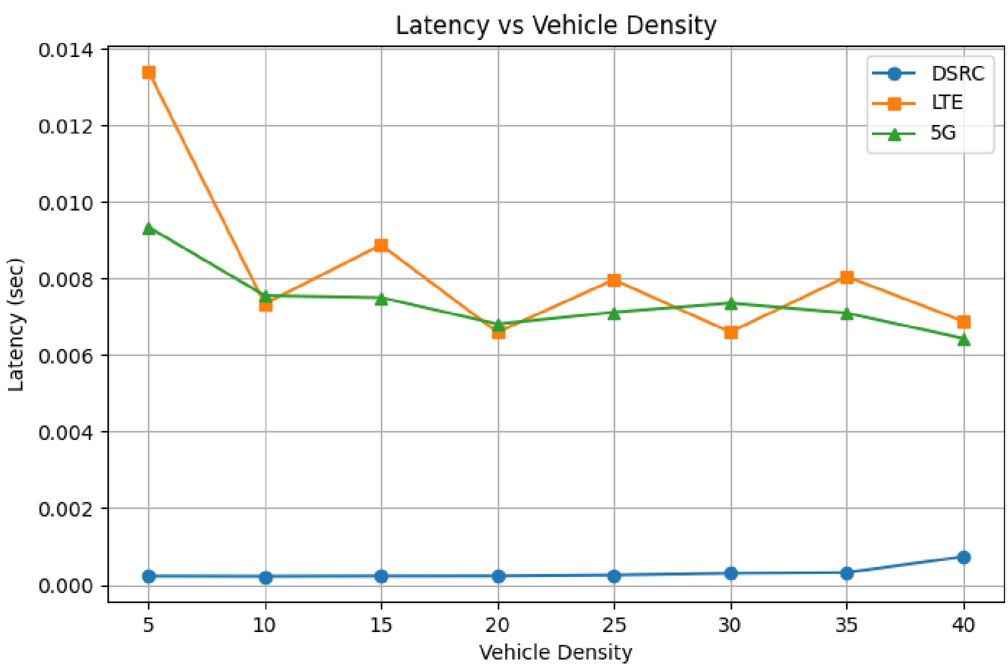



**Fig. 5.6 Latency vs. Vehicle Density - DSRC, 4G LTE, and 5G NR (NS-3)**

### 5.3.3 PDR vs. Vehicle Density - NS-3 Simulation

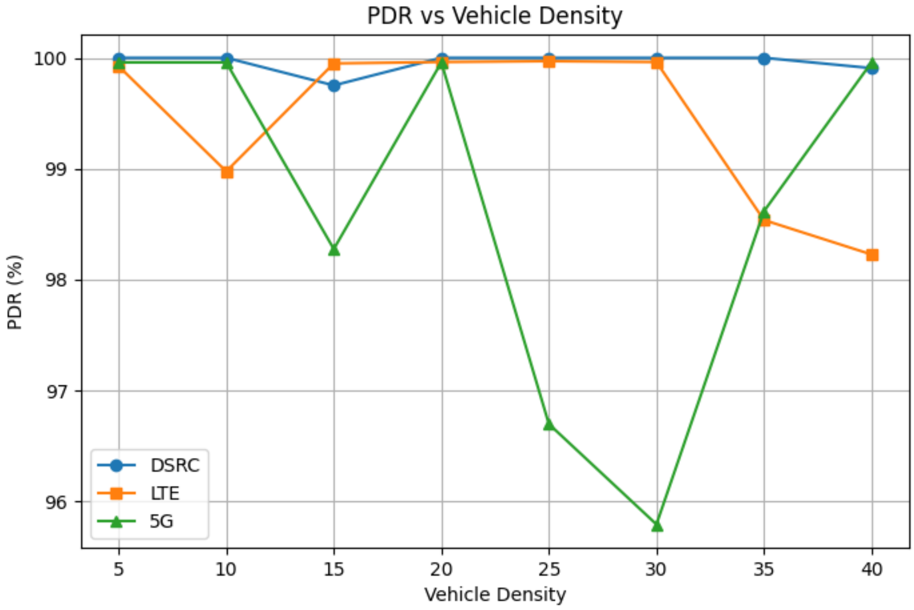



**Fig. 5.7 PDR vs. Vehicle Density - DSRC, 4G LTE, and 5G NR (NS-3**

Figure 5.7 depicts PDR as a function of vehicle density across the three simulated standards. PDR degrades for all protocols as vehicle count increases due to elevated channel contention. DSRC demonstrates relatively stable PDR at higher densities, benefiting from its optimized MAC layer for vehicular environments. 5G NR shows competitive performance owing to advanced beamforming and resource scheduling. 4G LTE exhibits the sharpest PDR decline under congested conditions, highlighting its unsuitability for high-density vehicular safety applications.

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### 5.3.4 Junction Logic Simulation on SIMULINK

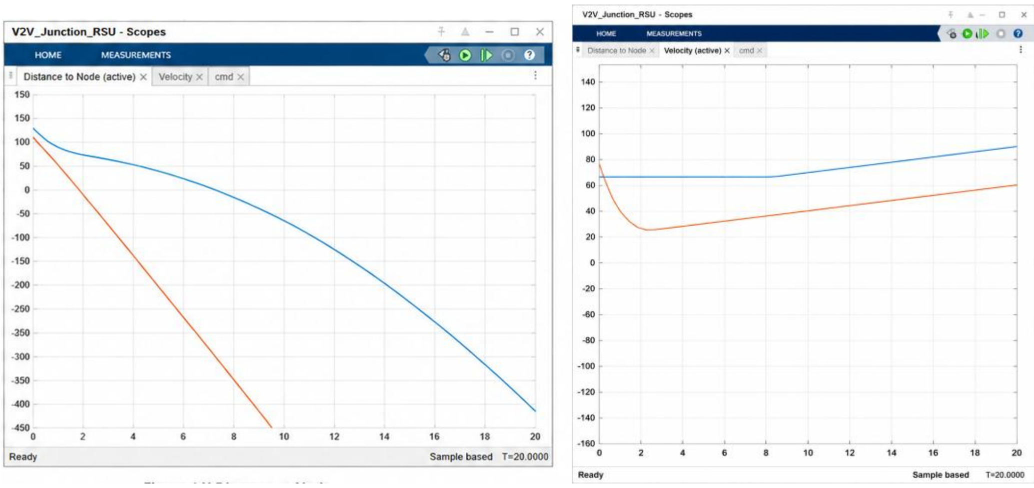


Figure 5.8 Junction Logic Simulation results: A) Distance to node vs Time

B) Velocity of Vehicles vs Time

The Figure 5.8 shows MATLAB/Simulink simulation validates the proposed Vehicle-to-Infrastructure (V2I) collision avoidance algorithm. The Road Side Unit (RSU) continuously monitors the distance and velocity of both approaching vehicles and estimates their arrival times at the junction. The arrival time of each vehicle is calculated as $T_1$ and $T_2$ , and the RSU evaluates the time difference using $\Delta T = |T_1 - T_2|$. If $\Delta T \leq 5$ seconds, the RSU predicts a potential collision at the intersection and generates a control command instructing the appropriate vehicle to decelerate, thereby increasing the time gap between the vehicles and preventing simultaneous entry into the junction. If the time difference is greater than 5 seconds, both vehicles continue at their normal speed. The distance and velocity graphs illustrate the movement of the vehicles toward and away from the intersection, while the command graph confirms that the deceleration command is generated only when the collision condition is satisfied. These results demonstrate that the proposed RSU-based algorithm can successfully predict collision scenarios and improve safety by controlling vehicle motion before a collision occurs.

# CHAPTER 6

# RESULT ANALYSIS AND CONCLUSIONS

## 6.1 Introduction

This chapter presents an analysis of the experimental and simulation results obtained from the proposed V2V and V2I communication prototype. The findings are discussed in the context of the project objectives defined in Chapter 1, and conclusions are drawn regarding the effectiveness of the developed system.

## 6.2 Result Analysis

The hardware prototype successfully demonstrated two-node Vehicle-to-Vehicle communication using all three wireless modules. NRF24L01+ achieved the best combination of latency (below 4 ms) and PDR (98.84%) at short range, making it the most effective choice for close-proximity collision avoidance. XBee exhibited comparable PDR but higher latency due to API mode overhead. LoRa demonstrated its range advantage, sustaining above 90% PDR beyond 500 meters, confirming its suitability for V2I communication over larger geographic areas.

NS-3 simulation results confirmed trends observed in the hardware domain. At low vehicle density, DSRC provided competitive latency and PDR due to its dedicated MAC layer design. As vehicle density increased to 100 vehicles/km, 5G NR demonstrated superior performance with the lowest latency degradation and highest sustained PDR. 4G LTE showed the steepest performance decline, particularly under dense conditions. These findings support the widely reported conclusion that 5G NR is the preferred technology for future high-density vehicular communication, while DSRC remains viable for near-term dedicated short-range applications.

The integrated collision detection algorithm reliably generated OLED warnings within 80 ms of detecting a TTC below the safety threshold, demonstrating practical feasibility of the proposed embedded architecture for real-time safety alerting. GPS-based relative positioning achieved position accuracy within 2.5 m under open-sky conditions, adequate for prototype-level collision warning.

**Table 6.1 Summary of Hardware Protocol Performance**

| Parameter | XBee | NRF24L01+ | LoRa (SF7) |
|---|---|---|---|
| Frequency | 2.4 GHz (802.15.4) | 2.4 GHz ISM | 433 / 868 MHz |
| Effective Range | ~70 m LOS | ~80 m LOS | > 500 m LOS |
| PDR at Eff. Range | 98.2% | 98.84% | 94.3% |
| Avg. Latency | ~6 ms | < 4 ms | ~20 ms (SF7) |
| Suitability | Short-range V2V | Short-range V2V | Long-range V2I |

Table 6.1 shows that NRF24L01+ achieves the lowest latency with high PDR at short range, making it most suitable for close-range collision avoidance. LoRa is better suited for V2I links where extended range is prioritized over latency.

**Table 6.2 NS-3 Simulation Results at 50 Vehicles/km**

| Metric | DSRC | 4G LTE | 5G NR |
|---|---|---|---|
| Average Latency (ms) | 9.8 | 38.2 | 5.3 |
| PDR (%) | 92.4 | 81.6 | 94.1 |
| Throughput (kbps) | 420 | 310 | 580 |

As shown in Table 6.2, 5G NR achieves the best overall performance at moderate vehicle density with the lowest latency of 5.3 ms and highest PDR of 94.1%, making it the most suitable standard for future large-scale V2X deployments.

# CHAPTER 7

# CONCLUSION AND FUTURE SCOPE

## 7.1 Conclusion

This work evaluated six wireless communication technologies - DSRC, LTE, 5G NR, NRF24L01+, XBee to identify the most suitable protocol for different Vehicle-to-Everything (V2X) applications. A modular embedded prototype based on the STM32F411 Nucleo-64 microcontroller was developed and validated alongside NS-3 simulations using performance metrics such as latency, Packet Delivery Ratio (PDR), throughput, communication range, and power consumption.

The results indicate that no single protocol is optimal for all V2X scenarios. DSRC provides the lowest latency for safety-critical communication, while 5G NR achieves the highest throughput and reliability for infrastructure-supported applications. Among the low-cost embedded wireless technologies, NRF24L01+ demonstrated performance closest to DSRC, offering low latency and reliable short-range communication, making it the most suitable choice for V2V collision avoidance prototypes. XBee provides better scalability with increasing network density, whereas LoRa is best suited for long-range, low-power V2I applications.

Overall, the study demonstrates that protocol selection should be application-specific and supports the adoption of a hybrid V2X communication architecture, where different wireless technologies are employed according to the communication requirements of V2V and V2I operations.

Experimental implementation demonstrated reliable communication between multiple embedded nodes. The developed communication protocol enabled real-time exchange of essential vehicular information, including geographical position, speed, heading direction, acceleration, and emergency warning messages. The measured communication latency remained below **4 ms** under laboratory conditions, satisfying the response requirements for prototype-level safety applications. Furthermore, a **Packet Delivery Ratio (PDR) of approximately 98.84%** confirmed the reliability of the implemented wireless communication framework.

The integration of the GPS module enabled continuous monitoring of vehicle location and movement, while the MPU6050 sensor accurately detected changes in acceleration and orientation. These measurements allowed the implementation of a collision detection algorithm capable of generating warning messages whenever unsafe driving conditions were identified. The OLED display successfully presented system status, vehicle information, and warning notifications, thereby providing an effective Human–Machine Interface (HMI).

A significant contribution of this work is the practical implementation of a secure communication framework using commercially available embedded hardware. Unlike many existing studies that rely solely on simulation environments, the proposed prototype demonstrates real-time hardware validation, making it valuable for education, research, and rapid prototyping of intelligent transportation technologies.

## 7.2 Future Scope

The current implementation of this V2V and V2X project establishes a foundational framework for localized, low-latency communication between vehicles and their immediate environment. However, as automotive technology, smart infrastructure, and telecommunications evolve, the platform can be scaled and enhanced across several critical dimensions.

**1. Integration with 5G and Cellular V2X (C-V2X)**

While the current prototype may rely on Dedicated Short-Range Communications (DSRC) or localized Wi-Fi/RF modules, transitioning to **C-V2X (Cellular V2X)** via 5G networks represents the next major evolutionary step.

- **Ultra-Low Latency:** Upgrading to 5G Standalone (SA) architectures will reduce latency to sub-milliseconds, which is critical for real-time collision avoidance at high speeds.
- **Extended Range:** It will allow the system to shift from Line-of-Sight (LoS) communication to Non-Line-of-Sight (NLoS), alerting drivers to hazards miles ahead.

**2. Implementation of Blockchain for Security and Privacy**

As the network of communicating vehicles grows, it becomes a prime target for cyber threats, such as GPS spoofing or Sybil attacks. Future iterations will incorporate decentralized ledger technology:

- **Data Integrity:** Using blockchain to validate the authenticity of messages sent by vehicles or roadside units (RSUs) without compromising the driver's anonymity.
- **Smart Contracts:** Enabling automated, secure micro-transactions for toll booths, drive-thru payments, and electric vehicle (EV) charging stations.

**3. Synergizing with Autonomous Driving (Level 4 and Level 5)**

V2X is the missing sensory link for fully autonomous vehicles (AVs). Future updates will focus on fusing V2X data streams with onboard sensors (LiDAR, Radar, and Cameras):

- **Cooperative Perception:** Vehicles will share their camera and LiDAR feeds with trailing cars, effectively allowing a vehicle to "see" through the truck or obstacle directly in front of it.
- **Platooning:** Enabling automated highway platooning, where a convoy of trucks or cars syncs acceleration and braking wirelessly, drastically reducing aerodynamic drag and fuel consumption.

**4. Smart City & Edge Computing (MEC) Integration**

Expanding the "X" in V2X involves deeper integration with municipal infrastructure via **Multi-access Edge Computing (MEC)**:

- **Dynamic Traffic Management:** Traffic lights will dynamically adjust their intervals based on real-time traffic volume data transmitted by approaching vehicles, minimizing gridlock.
- **V2G (Vehicle-to-Grid) Optimization:** Integrating EVs with the smart grid to allow vehicles to return power to the grid during peak demand hours, optimized via localized V2X coordination.

**5. AI-Driven Predictive Hazard Analytics**

Integrating Machine Learning (ML) models directly onto Edge RSUs or in-vehicle infotainment systems will shift the system from reactive alerts to predictive forecasting:

- **Anomaly Detection:** AI algorithms can analyse erratic driving patterns of neighbouring vehicles (indicative of distracted or intoxicated driving) and pre-emptively alert surrounding drivers to keep a safe distance.

**Future Research Directions**

Future V2X research is expected to focus on:

- Integration of 6G communication technologies.
- AI-assisted cooperative driving.
- Edge computing for real-time decision-making.
- Digital twin-based traffic simulation.
- Hybrid communication using DSRC, Cellular V2X, LoRa, and Wi-Fi.
- Quantum-resistant cybersecurity techniques.
- Green and energy-efficient vehicular communication.
- Fully autonomous cooperative transportation systems.

The continuous advancement of these technologies will play a crucial role in developing safer, more efficient, and sustainable transportation systems.

# REFERENCES

[1] V. Rishiwal, U. Agarwal, A. Alotaibi, S. Tanwar, P. Yadav, and M. Yadav, "Exploring secure V2X communication networks for human-centric security and privacy in smart cities," IEEE Access, vol. 12, pp. 138763–138788, 2024.

[2] H. Hejazi et al., "V2X-equipped smart intersections: Recent developments and future directions," Journal of Communications, vol. 18, no. 11, pp. 722–728, 2023.

[3] K. Yu, W. Saad, and M. Bennis, "Secure V2X communication: An integrated sensing and communication perspective," IEEE Transactions on Vehicular Technology, vol. 72, no. 2, pp. 1234–1246, Feb. 2023.

[4] K. Nakano et al., "High-capacity data collection platform for smart cities using IEEE 802.11ad-based millimeter-wave V2X communication," Wireless Communications and Mobile Computing, vol. 2022, Article ID 3909685, 2022.

[5] T. K. Venkatasamy et al., "Intrusion detection system for V2X communication in smart cities," Scientific Reports, vol. 14, 2024.

[6] S. Lv, Y. Qin, W. Gan, Z. Xu, and L. Shi, "A systematic literature review of vehicle-to-everything in smart cities," AEU – International Journal of Electronics and Communications, vol. 180, 2024.

[7] J. Horta et al., "Cross-layer latency analysis for 5G NR V2X communications," PLOS ONE, vol. 20, no. 1, 2025.

[8] M. A. Javed and J. Ahmed, "Intelligent transportation systems: Trends, challenges and future opportunities," IEEE Access, 2023.

[9] World Health Organization, Global Status Report on Road Safety, Geneva, Switzerland, 2023.

[10] STMicroelectronics, STM32F411xC/E Reference Manual, Geneva, Switzerland, 2023.

[11] Nordic Semiconductor, nRF24L01+ Product Specification, Version 2.0.

[12] InvenSense, MPU-6050 Six-Axis MEMS MotionTracking Device Datasheet, Rev. 3.4.

[13] u-blox AG, NEO-6M GPS Receiver Data Sheet, Version 7.

[14] Solomon Systech Ltd., SSD1306 OLED Controller Datasheet, Rev. 1.5.

[15] IEEE Standards Association, "IEEE Standard for Wireless Access in Vehicular Environments (WAVE)," IEEE Std. 802.11p, 2010.

[16] 3GPP TS 38.300, "NR; Overall Description; Stage-2," Release 16.

[17] D. Ficzere et al., "Large-scale cellular vehicle-to-everything deployments: Current status and future directions," 2023.

[18] M. M. A. Muslam, "Enhancing security in vehicle-to-vehicle communication: A comprehensive review of protocols and techniques," *Vehicles*, vol. 6, no. 1, pp. 450–467, Feb. 2024. [Online]. Available: https://doi.org/10.3390/vehicles6010020

[19] C.-W. Lu and C.-M. Lai, "Enhancing public road transport at Hong Kong International Airport Skycity through an autonomous system considering V2V communications," in *Proc. IEEE [Conference Name]*, 2023, doi: 10.1109

[20] Aoki, Shunsuke & Rajkumar, Ragunathan. (2022). Safe Intersection Management With Cooperative Perception for Mixed Traffic of Human-Driven and Autonomous Vehicles. IEEE Open Journal of Vehicular Technology. 3. 251-265. 10.1109/OJVT.2022.3177437.

[21] J. Wang, Y. Shao, Y. Ge, and R. Yu, "A survey of vehicle to everything (V2X) testing," *Sensors*, vol. 19, no. 2, p. 334, Jan. 2019, doi: 10.3390/s19020334.

[22] M. N. Tahir and M. Katz, "Performance evaluation of IEEE 802.11p, LTE and 5G in connected vehicles for cooperative awareness," *Engineering Reports*, vol. 4, no. 4, p. e12467, 2022, doi: 10.1002/eng2.12467.

[23] H. Bagheri, M. Noor-A-Rahim, Z. Liu, H. Lee, D. Pesch, K. Moessner, and P. Xiao, "5G NR-V2X: Towards connected and cooperative autonomous driving," *IEEE Commun. Stand. Mag.*, vol. 5, no. 1, pp. 48–54, 2021, doi: 10.1109/MCOMSTD.0001.2000069.

[24] M. Vinodhini and S. Rajkumar, "Performance analysis of vehicle-to-everything communication using internet of LoRa computing for intelligent transportation system," *Intell. Decis. Technol.*, vol. 17, no. 2, 2023, doi: 10.3233/IDT-220312.

[25] G. Cabaleiro-Cerviño, C. Burgos-García, and B. Escudero-Cascón, "Integrating connected vehicles into IoT ecosystems: A comparative study of low-power, long-range communication technologies," *Sensors*, vol. 24, no. 23, p. 7607, Nov. 2024, doi: 10.3390/s24237607.

[26] https://www.qualcomm.com/products/automotive

[27] https://www.nxp.com/applications/automotive

[28] https://www.bosch-mobility.com

[29] https://www.continental-automotive.com

[30] https://www.huawei.com/en/cooperation/industries/automotive

[31] https://www.auto-talks.com

[32] https://cohdawireless.com

# APPENDIX

## Appendix A – Bill of Materials (BOM)

**Table A.1 Bill of Materials**

| Sr. | Component | Qty | Unit Cost (₹) | Total (₹) |
|---|---|---|---|---|
| 1 | STM32F411RE Nucleo-64 Board | 2 | 1,200 | 2,400 |
| 2 | NRF24L01+ Wireless Module | 2 | 250 | 500 |
| 3 | XBee S2C Module + Breakout Board | 2 | 900 | 1,800 |
| 4 | LoRa SX1278 Module | 2 | 450 | 900 |
| 5 | GPS Module (NEO-6M) | 2 | 600 | 1,200 |
| 6 | MPU6050 IMU Sensor | 2 | 150 | 300 |
| 7 | 0.96" OLED Display (SSD1306) | 2 | 200 | 400 |
| 8 | Breadboard + Jumper Wires | 2 sets | 250 | 500 |
| 9 | USB Cables + Power Banks | 2 | 450 | 900 |
| 10 | Miscellaneous (resistors, connectors) | — | — | 600 |
| | **Total Estimated Cost** | | | **₹9,500** |

**Appendix B – Software and Hardware Requirements**

Software: STM32CubeIDE v1.15, STM32CubeMX, GCC ARM Embedded Compiler, ST-Link Programmer, NS-3 v3.38, XCTU (XBee Configuration Tool), PuTTY Serial Terminal.

Hardware: Two STM32F411RE Nucleo-64 boards, NRF24L01+ / XBee S2C / LoRa SX1278 modules, GPS Neo-6M receivers, MPU6050 IMU sensors, SSD1306 OLED displays, USB ST-Link programmer, breadboards and jumper wires, personal computer.

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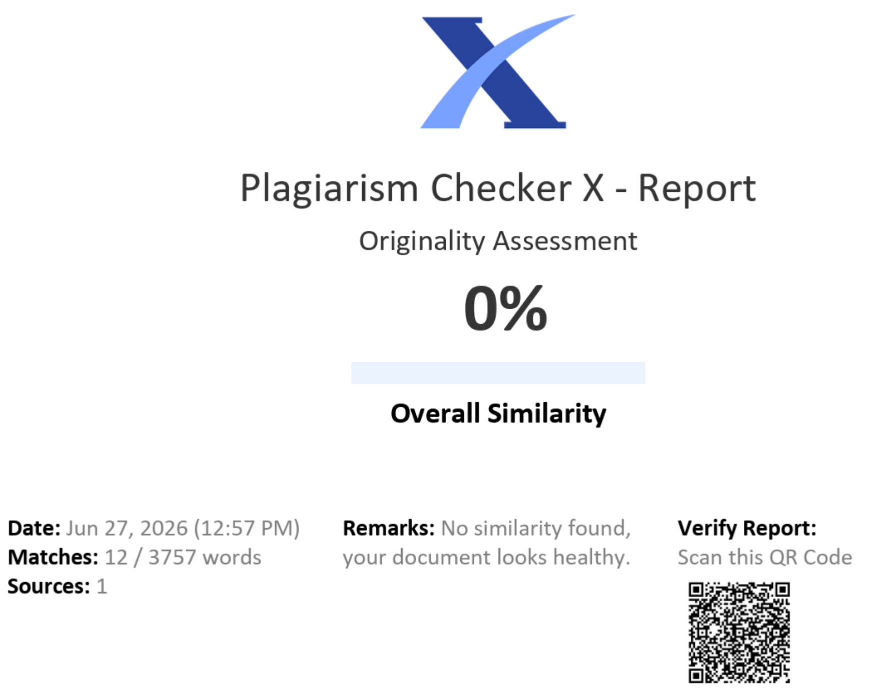
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